From Black-Box Models to Transparent Rules: A RuleFit Approach to Credit Risk Assessment

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Abstract. This report investigates the trade-off between predictive performance and interpretability in credit risk assessment using the German Credit dataset. The study begins with data profiling and preprocessing, including the treatment of missing values, categorical encoding, and feature transformation, in order to obtain a machine-learning-ready representation of the problem. Several classification models are then trained and compared, including Gaussian Naive Bayes, Random Forest, XGBoost, and a hybrid pipeline combining dimensionality reduction, feature selection, and Naive Bayes. Their performance is evaluated with special attention to F1-score, as a balanced metric that jointly considers precision and recall in the context of credit risk classification.
Beyond predictive modelling, the report focuses on interpretable rule-based analysis through RuleFit. First, RuleFit is trained directly on the credit risk labels as an interpretable classifier, allowing the extraction of compact human-readable rules and the identification of the most influential variables in the decision process. Second, RuleFit is used as a surrogate model to distill the behaviour of the selected teacher models, providing a rule-based approximation of their decision logic and enabling a fidelity-based comparison between teachers and surrogates.
The results show that, while black-box models achieve the strongest predictive performance, RuleFit offers a valuable complementary perspective by translating complex decision patterns into transparent symbolic rules. Overall, the findings support its usefulness as a practical tool for explainable credit scoring.

Keywords: Rule extraction · Explainable AI · RuleFit · Credit risk

1 Introduction

Credit risk assessment is a fundamental task in the financial domain, as it supports decisions about whether a loan applicant should be classified as a reliable or risky borrower. Errors in this setting may have important economic consequences: approving high-risk applicants can increase default losses, whereas rejecting creditworthy applicants may reduce business opportunities and fairness in access to financial services. For this reason, credit scoring has long been

studied as a supervised classification problem, and benchmark datasets such as the *Statlog (German Credit Data)* dataset have become standard testbeds for evaluating predictive models in this context [1, **?**].

In recent years, machine learning models have significantly improved predictive performance in credit scoring, especially through the use of more expressive methods such as ensemble models and neural networks. However, these gains often come at the cost of reduced transparency. In high-stakes domains such as finance, predictive accuracy alone is not sufficient: model decisions must also be understandable, auditable, and justifiable. This has motivated the growing relevance of Explainable Artificial Intelligence (XAI), whose purpose is to make complex predictive systems more transparent and interpretable. Prior work in credit scoring has explicitly highlighted the tension between predictive performance and comprehensibility, and has shown that rule extraction and local explanation methods can play a key role in bridging this gap [2, 5, 7].

This report addresses that trade-off by combining predictive modelling with rule-based interpretability on the German Credit dataset. First, the data are profiled and preprocessed in order to obtain a suitable representation for machine learning. Then, several predictive models are trained and compared, including Gaussian Naive Bayes, Random Forest, XGBoost, and a hybrid pipeline based on dimensionality reduction, feature selection, and Naive Bayes. Finally, the study focuses on RuleFit in two complementary ways: as an interpretable classifier trained directly on the true labels, and as a surrogate model used to distill the behaviour of stronger black-box models into compact human-readable decision rules. In this way, the report does not only evaluate predictive performance, but also studies how complex model behaviour can be translated into transparent symbolic representations that are more suitable for explanation and inspection [2, 5].

2 Dataset and Methodology

2.1 Data Composition

This study is based on the *Statlog (German Credit Data)* dataset, a widely used benchmark for credit scoring and binary classification problems [1]. The dataset contains 1,000 loan applicants described by a set of financial and demographic attributes, with the objective of predicting whether each individual should be classified as a good or bad credit risk. These attributes include numerical variables such as age, loan duration, and credit amount, as well as categorical descriptors related to savings accounts, checking accounts, housing status, employment profile, and loan purpose. The target variable is binary and presents a moderately imbalanced distribution, with a larger proportion of good credit cases than bad ones, which reflects the type of asymmetry commonly found in real credit decision scenarios [2, 7].

The use of this dataset is particularly appropriate for the objectives of this report for two main reasons. First, German Credit has become a standard bench-

mark in the literature on credit scoring, financial risk modelling, and interpretable machine learning, making it a suitable basis for comparison with previous studies [4, 5]. Second, credit approval is a high-stakes application domain in which both predictive performance and interpretability are essential. Models must be able not only to identify risky applicants with sufficient accuracy, but also to provide decision logic that can be inspected, justified, and communicated to domain experts or auditors [2, 6].

2.2 Preprocessing Pipeline

Before model training, the dataset was subjected to a preprocessing stage aimed at obtaining a clean and machine-learning-ready representation. Missing values in the categorical variables *Saving accounts* and *Checking account* were explicitly handled by introducing a new category (`no_inf`), allowing the absence of information itself to be treated as a potentially informative signal rather than discarding observations. Subsequently, categorical variables such as loan purpose, sex, housing status, saving accounts, checking accounts, and age category were transformed into binary indicators through one-hot encoding. Once the encoded variables were created, the original categorical columns were removed to avoid redundancy in the feature space.

In addition, the variable *Credit amount* was log-transformed in order to reduce skewness and mitigate the influence of extreme values. This transformation is common in financial datasets, where monetary quantities often present long-tailed distributions. After preprocessing, the binary target variable was defined as `Risk_bad`, while the remaining columns were used as predictive features. Finally, the dataset was split into training and test subsets using a 75%–25% partition, which allowed all predictive and interpretability analyses to be evaluated on unseen data.

2.3 Predictive Modelling Strategy

To establish predictive baselines, the methodology first considered a broad exploratory comparison across several classification algorithms, including Logistic Regression, Linear Discriminant Analysis, K-Nearest Neighbors, Decision Trees, Gaussian Naive Bayes, Random Forest, Support Vector Machines, and XGBoost. This initial benchmark was used as a lightweight model screening stage, allowing different model families to be assessed before moving to more detailed tuning and interpretation.

Special attention was given to *F1-score* as a central evaluation metric. In the context of credit risk, both false negatives and false positives are relevant: failing to identify a truly risky applicant may lead to direct financial losses, while incorrectly rejecting a creditworthy applicant may reduce business opportunities and negatively affect fairness in access to credit. For this reason, F1-score is especially informative, since it provides a balanced summary of precision and recall and is therefore well suited to problems where performance on both classes must be considered jointly. Although additional metrics such as accuracy, recall,

precision, and confusion matrices were also reported, F1-score played a key role in the comparative evaluation and in the interpretation of model suitability for the task.

After this initial benchmark, four predictive approaches were selected for deeper analysis: Gaussian Naive Bayes, Random Forest, XGBoost, and a hybrid pipeline combining Principal Component Analysis (PCA), univariate feature selection, and Gaussian Naive Bayes. Gaussian Naive Bayes provides a simple probabilistic baseline, while Random Forest and XGBoost represent stronger tree-based ensemble methods capable of modelling complex non-linear interactions [4, 8]. The hybrid pipeline was included as an additional experimental setting in which feature reduction and feature selection were combined before classification. Hyperparameter tuning was performed for the main candidate models in order to obtain competitive configurations before proceeding to interpretability analysis.

2.4 RuleFit for Interpretable Modelling

Beyond predictive performance, the central methodological contribution of this work lies in the use of RuleFit as an interpretable rule-based learning technique. RuleFit, originally proposed by Friedman and Popescu, combines the expressive power of tree-based rule generation with the sparsity of regularized linear models [3]. The method first generates decision rules from tree structures and then fits a sparse linear model over those rules, optionally together with the original linear features. The resulting model retains only the most relevant terms, producing a compact set of human-readable conditions that can be directly inspected.

In this report, RuleFit is employed in two complementary ways. First, it is trained directly on the true class labels in order to evaluate it as an interpretable classifier in its own right. This allows us to assess whether meaningful and concise rules can be learned directly from the German Credit dataset while maintaining competitive predictive behaviour. Second, RuleFit is used as a *surrogate model* for rule distillation: instead of learning from the ground-truth labels, it is trained on the predictions generated by stronger teacher models. In this setting, RuleFit is not intended to outperform the teacher, but rather to approximate its behaviour through symbolic rules.

2.5 Rule Distillation from Teacher Models

The final methodological stage consists of using RuleFit to distill the behaviour of the selected predictive models. For each teacher model, the training predictions are used as surrogate targets, and a new RuleFit classifier is trained to reproduce that decision logic. The resulting rule sets are then evaluated according to two complementary criteria: predictive performance with respect to the true test labels, and *fidelity* with respect to the original teacher outputs. Fidelity measures the degree to which the distilled RuleFit model matches the decisions of the teacher, and therefore quantifies how well the black-box behaviour can be approximated by a transparent symbolic representation.

This design makes it possible to analyze not only which models perform well, but also which of them are more amenable to rule-based approximation. In that sense, the methodology goes beyond standard model evaluation and addresses a central XAI question: whether the predictive logic of strong black-box systems can be transferred into a compact and auditable set of rules without an excessive loss of information. This perspective is especially relevant in credit scoring, where interpretability is not merely desirable, but often necessary for trust, auditability, and practical deployment [2, 5, 6].

3 Results and Analysis

This section summarizes the most relevant empirical results obtained in the notebook. The focus here is descriptive: we report the predictive performance of the selected models, present the main outcomes of the RuleFit-based interpretation, and summarize the rule-distillation experiments carried out on top of the teacher models. A more detailed interpretation of why these results arise, and what they imply from a methodological perspective, is deferred to the discussion section.

3.1 Predictive performance of the selected models

After the initial exploratory benchmark, four models were retained for deeper analysis: Gaussian Naive Bayes, Random Forest, XGBoost, and a hybrid pipeline combining PCA, feature selection, and Gaussian Naive Bayes. In the final experimental setting, the best overall predictive accuracy on the test set was obtained by XGBoost, with an accuracy of 0.748, followed by the PCA+KBest+GaussianNB pipeline with 0.720, Random Forest with 0.712, and GaussianNB with 0.648. These values correspond to the teacher models later used in the rule-distillation stage and provide a compact summary of the final predictive baselines achieved in the notebook.

A concise way of reporting these results is shown in Table 1. The table is intentionally limited to the final selected models, since the broader cross-validation benchmark served mainly as a preliminary screening stage rather than as the main result of the notebook.

Table 1. Predictive accuracy of the selected models on the test set.

Model	Accuracy
GaussianNB	0.648
Random Forest	0.712
PCA+KBest+GaussianNB	0.720
XGBoost	0.748

These results already suggest that the tree-based ensemble methods, and especially XGBoost, provide the strongest predictive baseline among the models

considered, while simpler probabilistic methods remain useful as reference points. At the same time, the differences are not extreme, which makes the dataset suitable for studying the trade-off between performance and interpretability in a controlled manner.

3.2 RuleFit as an interpretable classifier

RuleFit was first trained directly on the true target labels in order to evaluate it as an interpretable classifier. On the test set, RuleFit achieved an accuracy of 0.7400, precision of 0.6522, recall of 0.2083, F1-score of 0.3158, and ROC-AUC of 0.7619. The associated confusion matrix was

$$\begin{bmatrix} 170 & 8 \\ 57 & 15 \end{bmatrix},$$

which indicates that the model was quite conservative when assigning the positive class, producing relatively few false positives but also missing a large number of bad-credit cases.

These results show that RuleFit remained competitive in terms of overall accuracy and ROC-AUC, but its recall for the minority class was clearly lower than what would be desirable in a risk-sensitive setting. This behaviour is important because it illustrates an early manifestation of the main trade-off explored throughout the notebook: the interpretability gain offered by sparse rules does not automatically translate into stronger sensitivity to the bad-credit class.

The extracted rules nevertheless provide a meaningful symbolic summary of the task. Among the most influential conditions, the model identified patterns involving *Credit amount*, *Duration*, *Age*, *Checking account*, *Savings*, and housing-related variables. Examples of high-impact rules include:

- `Age > 35.5 and Credit amount <= 8.26769` with coefficient -0.84,
- `Savings_no_inf <= 0.5 and Check_no_inf <= 0.5 and Duration > 22.5` with coefficient 0.54,
- `Check_no_inf <= 0.5 and Credit amount <= 7.68789` with coefficient 0.50.

These examples illustrate that RuleFit captured non-linear interactions between financial profile variables rather than relying only on isolated single-feature effects.

3.3 Rule distillation from teacher models

The final stage of the notebook used RuleFit as a surrogate model to approximate the behaviour of the four selected predictive approaches. For each teacher, the surrogate was trained on the teacher predictions rather than on the true labels, and the resulting models were evaluated both in terms of predictive accuracy and in terms of fidelity to the original teacher.

The results are summarized in Table 2. In terms of fidelity, the easiest models to approximate were the PCA+KBest+GaussianNB pipeline and GaussianNB,

Table 2. RuleFit surrogate results for the four teacher models.

Teacher	Teacher Acc.	Surrogate Acc.	Fidelity	# Rules
GaussianNB	0.648	0.668	0.956	34
Random Forest	0.712	0.748	0.828	40
PCA+KBest+GaussianNB	0.720	0.708	0.988	36
XGBoost	0.748	0.724	0.848	32

with fidelity values of 0.988 and 0.956, respectively. XGBoost and Random Forest were harder to approximate, with lower fidelities of 0.848 and 0.828. However, the most accurate distilled surrogate on the true test labels was the one learned from Random Forest, which reached an accuracy of 0.748, followed by the XGBoost surrogate with 0.724 and the PCA+KBest+GaussianNB surrogate with 0.708.

A second relevant observation concerns the size of the extracted rule sets. The distilled RuleFit models remained relatively compact, ranging from 32 to 40 active rules depending on the teacher model. This suggests that, even when fidelity is not perfect, the surrogate models can still provide a relatively concise symbolic approximation of the original decision logic.

The fidelity ordering is also visually clear in Fig. 1. The two simpler teacher models, GaussianNB and PCA+KBest+GaussianNB, are approximated with very high fidelity, whereas the ensemble-based teachers, Random Forest and XGBoost, show noticeably lower agreement with their RuleFit surrogates. This suggests that simpler or more regular decision boundaries are easier to compress into sparse rule sets than the more complex behaviour of ensemble models.

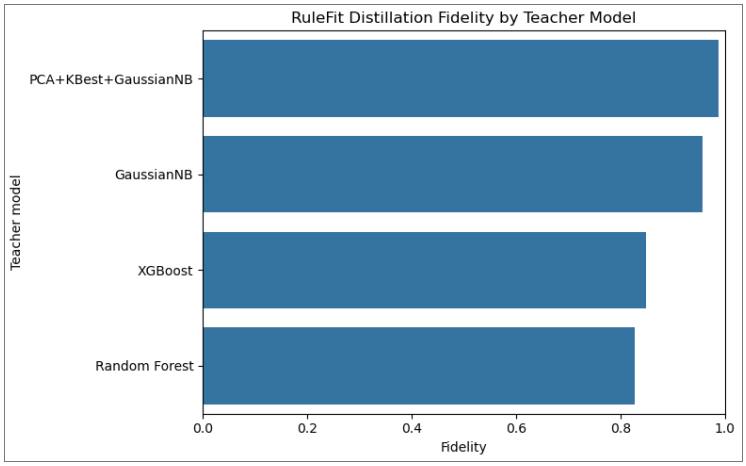

Fig. 1. Fidelity of the RuleFit surrogates with respect to the teacher models.

4 Discussion

The results obtained in this work reinforce the central trade-off that motivated the study: in credit risk assessment, the strongest predictive models are often also the least transparent. Among the four selected approaches, XGBoost achieved the highest test accuracy, while Random Forest and the PCA+KBest+GaussianNB pipeline also provided competitive results. This confirms that, even on a relatively small benchmark such as German Credit, ensemble and hybrid approaches are able to capture relevant decision patterns that go beyond those learned by simpler probabilistic baselines. At the same time, the differences in accuracy across models were not extreme, which makes the dataset particularly suitable for studying whether interpretability can be introduced without an excessive loss of predictive utility.

When RuleFit was trained directly on the true labels, it produced a model that remained competitive in terms of overall accuracy and ROC-AUC, while also yielding explicit decision rules involving financially meaningful variables such as *Credit amount*, *Duration*, *Age*, and the status of checking and savings accounts. This is an important result, since it shows that a rule-based model can recover non-linear interactions that are relevant for credit scoring while still remaining human-readable. However, the relatively low recall obtained by RuleFit on the minority class also indicates that interpretability does not automatically imply balanced performance. In this sense, the direct RuleFit model provided a transparent representation of the task, but not necessarily the best sensitivity to risky applicants. This behaviour is consistent with the broader literature on interpretable credit scoring, where the gain in comprehensibility often comes with some loss in flexibility or discriminatory power [2, 5, 3].

The surrogate experiments provide a second, complementary perspective. When RuleFit was trained to imitate the selected teacher models, it achieved high fidelity for the simpler GaussianNB-based teachers and lower, but still meaningful, fidelity for Random Forest and XGBoost. This pattern is intuitively plausible: simpler decision functions are easier to compress into sparse symbolic rules, whereas ensemble methods rely on more fragmented and distributed decision boundaries that are harder to approximate perfectly with a compact rule set. From an XAI perspective, this suggests that not all black-box models are equally amenable to rule-based distillation. Some models can be approximated very closely without losing much information, while others preserve part of their predictive advantage precisely because their internal behaviour is more complex [3, 6].

A particularly interesting observation is that the distilled RuleFit surrogate of Random Forest achieved a test accuracy slightly above that of the original teacher. This should not be interpreted as evidence that the surrogate is intrinsically superior to the black-box model, but rather as an indication that the distilled rule set may, in some cases, act as a simplified and regularized approximation of the teacher that generalizes differently on the test set. In other words, the surrogate is not only copying the original decision logic, but also imposing a sparsity constraint that can suppress noisy or overly specific patterns.

In contrast, the surrogate of XGBoost preserved relatively high fidelity while remaining slightly below the teacher in predictive performance, which is more closely aligned with the usual expectation for surrogate modelling.

From an application perspective, the findings are especially relevant for credit scoring. In financial decision-making, interpretability is not merely a desirable extra, but often a practical requirement for trust, auditability, and justification. The fact that RuleFit was able both to operate as an interpretable classifier and to distill the logic of stronger models into explicit rules makes it a valuable tool in this domain. Rather than treating interpretability and predictive performance as mutually exclusive, the notebook results suggest that they can be combined in complementary ways: black-box models can be used to maximize predictive strength, while RuleFit can be used either to provide a directly interpretable alternative or to generate a transparent symbolic approximation of the original decision process.

At the same time, several limitations should be acknowledged. First, the experiments are based on a single benchmark dataset of moderate size, so the conclusions should not be generalized too broadly without additional validation on larger or more heterogeneous credit datasets. Second, the extracted rules are necessarily influenced by the preprocessing choices and by the hyperparameters of both the teacher models and RuleFit itself. Third, the evaluation focused mainly on predictive metrics and fidelity, while other aspects that are highly relevant in financial AI, such as robustness, fairness, and the stability of the extracted rules across different data splits, were not explored in full depth. These limitations do not weaken the main contribution of the work, but they do define a natural path for future extensions.

5 Conclusions

This report has investigated the relationship between predictive performance and interpretability in credit risk assessment using the German Credit dataset as an applied benchmark. The experimental workflow combined data profiling, preprocessing, predictive modelling, and rule-based interpretability analysis in order to study not only how accurately different models classify credit applicants, but also how transparently their decision logic can be represented.

The predictive results showed that XGBoost achieved the strongest overall performance among the selected baselines, with Random Forest and the PCA+KBest+GaussianNB pipeline also obtaining competitive results. These models confirmed the usefulness of ensemble and hybrid approaches for capturing non-linear patterns in the data. At the same time, the experiments also showed that simpler models, although generally weaker in predictive terms, remain valuable reference points for understanding the behaviour of more complex approaches.

A central contribution of the report was the use of RuleFit in two complementary ways. First, RuleFit was evaluated as an interpretable classifier trained directly on the true target labels. In this setting, it was able to extract con-

cise and human-readable rules involving financially meaningful variables such as credit amount, duration, age, and account status, while maintaining competitive overall predictive performance. Second, RuleFit was used as a surrogate model for rule distillation, approximating the behaviour of the previously trained predictive models through sparse symbolic rules. These distillation experiments showed that RuleFit can achieve high fidelity for simpler teacher models and still provide meaningful approximations for stronger black-box models such as Random Forest and XGBoost.

Taken together, the results support the idea that RuleFit is a useful bridge between accuracy and transparency in credit scoring. While interpretable models do not always match the predictive strength of the best black-box baselines, they can still offer an informative representation of the main decision patterns learned from the data. In highly regulated and high-stakes domains such as finance, this capacity to turn predictive logic into explicit rules is especially valuable.

As future work, it would be interesting to extend the analysis to larger credit scoring datasets, evaluate the stability of the extracted rules across multiple data splits, and incorporate additional dimensions such as fairness, robustness, and local explanation. These extensions would provide a broader view of how rule-based interpretability methods can support trustworthy machine learning in financial applications.

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